3. $y''(t) - y'(t) - 6y(t) = t^2 + 3t$ partikularno

$y_p = At^2 + Bt + C$

$y_p' = 2At + B \qquad y_p'' = 2A$

$2A - 2At + B - 6At^2 - 6Bt - 6C = t^2 + 3t$

$-6A = 1 \rightarrow A = \frac{-1}{6}$

$-2A - 6B = 3$

$\frac{1}{3} - 6B = 3 \rightarrow B = \frac{8}{18} = \frac{4}{9}$

$2A + B - 6C = 0$

$\frac{-1}{3} + \frac{4}{9} - 6C = 0$

$\frac{1}{9} = 6C \rightarrow C = \frac{1}{54}$

5. Partikularno: $y[n+2] + 2y[n+1] + y[n] = (-1)^n$

$y_p = K(-1)^n \qquad K(-1)^{n+2} + 2K(-1)^{n+1} + K(-1)^n = (-1)^n$

$K(-1)^2 + 2K(-1) + K = 1$

$K - 2K + K = 1 \rightarrow 0 = 1$

$y_p = Kn(-1)^n$

9. $g^2 + 5g + 6 = 0 \qquad g_1 = -2, \; g_2 = -3$

10. $y''(t) + 3y'(t) + 2y(t) = u'(t)$

$s^2 + 3s + 2 = 0 \qquad s_1 = -1, \; s_2 = -2$

$h_a(t) = C_1 e^{-t} + C_2 e^{-2t}, \; t \geq 0$

$h_a'(t) = -C_1 e^{-t} - 2C_2 e^{-2t}$

$C_1 + C_2 = 0$

$-C_1 - 2C_2 = 1$

$C_2 - 2C_2 = 1$

$C_2 = -1$

1. $6y[n+2] + 5y[n+1] + y[n] = 8u[n+2] + 4u[n]$

$6g^2 + 5g + 1 = 0 \qquad g_1 = \frac{-1}{3}, \; g_2 = \frac{-1}{2}$

2. $y'(t) + 3y(t) = u(t) \; ; \; u(t) = (e^{-3t} + e^{-4t})\mu(t), \; y(0^-) = 5$

totalni odziv = ? $\rightarrow y_h \Rightarrow s = -3 \rightarrow y_h = C \cdot e^{-3t}$

$y_{p_1} = K_1 e^{-3t} \cdot t \rightarrow y_{p_1}' = K_1 e^{-3t} - 3K_1 e^{-3t} \cdot t$

$y_{p_2} = K_2 e^{-4t} \rightarrow y_{p_2}' = K_2(-4)e^{-4t}$

$K_1 e^{-3t} - 3K_1 e^{-3t} \cdot t - 4K_2 e^{-4t} + 3K_1 e^{-3t} \cdot t + 3K_2 e^{-4t} = e^{-3t} + e^{-4t}$

$K_1 \cdot \cancel{e^{-3t}} - 3K_1 \cancel{e^{-3t}} \cdot t + 3K_1 \cancel{e^{-3t}} = \cancel{e^{-3t}}$

$4K_1 - 3K_1 \cdot t = 1$

$K_1 = \frac{1}{4}$

$-4K_2 \cancel{e^{-4t}} + 3K_2 \cancel{e^{-4t}} = \cancel{e^{-4t}}$

$-K_2 = 1 \rightarrow K_2 = -1$

$y(t) = C \cdot e^{-3t} + \frac{1}{4} e^{-3t} \cdot t - e^{-4t}$

$y(0) = 5 = C_1 + 0 - 1 \rightarrow C_1 = 6$

$y(t) = 6e^{-3t} + \frac{1}{4} t e^{-3t} - e^{-4t}$

6 $y'(t) + 2y(t) = u(t)$ $\quad b_0 = 0 \quad b_1 = 1$ $\quad u = \delta(t)$

$h_a'(t) + 2h_a(t) = u(t) \rightarrow s + 2 = 0 \rightarrow \underline{h_a(t) = C \cdot e^{-2t}}$

$\underline{h_a'(t) = -2Ce^{-2t}}$ $\quad C = 1 \quad \rightarrow \underline{h_a(t) = e^{-2t}}$

$$\boxed{h(t) = e^{-2t}\,\mu(t)}$$

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1 $5y[n-2] + 15y[n-1] + 5y[n] = 13u[n]$, prva tri uzorka odziva neprekinutog sustava za korake $n=0$, $n=1$, $n=2$.

$y(-2) = 0$, $y(-1) = 1$.

$5y[n-2] + 15y[n-1] + 5y[n] = 0$ $\quad n = 0$

$\cancel{5y[-2]} + 15y[-1] + 5y[0] = 0$

$15 + 5y[0] \rightarrow y[0] = -3$

$n = 1$

$5y[-1] + 15y[0] + 5y[1] = 0$

$5 - 45 + 5y[1] = 0 \longrightarrow 5y[0] = 40 \Rightarrow y[1] = 8$

$n = 2$ $\quad 5y[0] + 15y[1] + 5y[2] = 0$

$-15 + 15 \cdot 8 + 5y[2] = 0$

$15 \cdot 7 + 5y[2] = 0$

$3 \cdot 7 \longrightarrow -21 = y[2]$

4 $y'(t) + y(t) = \delta(t)$ $\quad b_0 = 0, \; b_1 = 1$

$s = -1 \rightarrow h_a(t) = Ce^{-t}$

$h_a'(t) = -Ce^{-t}$

$\alpha^2 + 3\alpha + 2 = 0 \qquad \alpha_1 = -1 \qquad \alpha_2 = -2 \qquad y_h[n] = C_1(-1)^n + C_2(-2)^n$

9. $s^2 - s - 6 = 0 \qquad s_1 = 3, \qquad s_2 = -2 \qquad y_h = C_1 e^{3t} + C_2 e^{-2t}$

1. $y''(t)+3y'(t)+2y(t)=u(t)$ $\quad h(t)=?$

$u(t)=\delta(t)$

$s^2+3s+2=0$ $\quad s_1=-1$ $\quad s_2=-2$ $\Rightarrow$ $\boxed{h_a(t)=C_1e^{-t}+C_2e^{-2t}}$

$h_a'(t)=-C_1e^{-t}-2C_2e^{-2t}$ $\quad h(0^-)=0$ $\quad h'(0^+)=1$

$$\left.\begin{aligned} C_1+C_2&=0\\ -C_1-2C_2&=1\end{aligned}\right\} \quad C_1=1 \quad C_2=-1 \quad h(t)=\left(e^{-t}-e^{-2t}\right)\mu(t)$$

2. $y[n]+11y[n-3]=5\delta[n]+4\delta[n-1]+2\delta[n-2]$ $\quad$ MIRNI SUSTAV!

$n=0$ $\quad y[0]+\underbrace{11y[-3]}_{0}=5 \rightarrow y[0]=5$ $\quad$ poć unazad ($<\emptyset$) su $\emptyset$!

$n=1$ $\quad y[1]+\underbrace{11y[n-2]}_{0}=\emptyset+4+\emptyset \rightarrow y[1]=4$

3. $y[n+2]+3y[n+1]+2y[n]=(-1)^n$

$q^2+3q+2=0$ $\quad q_1=-1$ $\quad q_2=-2$ $\Rightarrow$ $y_h[n]=C_1(-1)^n+C_2(-2)^n$

5. $3y'(t)+2y(t)=0{,}3\,\mu(t)$ $\quad$ part:

$3s=-2 \rightarrow s=\frac{-2}{3} \Rightarrow y_h(t)=Ce^{-\frac{2}{3}t}$

$y_p=K$

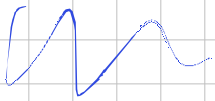
$y_p'=0$

$0+2K=0{,}3$

$K=\frac{3}{20}=\frac{15}{100}$

$y(t)=Ce^{-\frac{2}{3}t}+0{,}15$

Prava DZ. 8.

1

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